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(6) Aim: To Calculate the total cost of a purchase, including discount & tax, using arithmatic operations

Software Used:

Python 3 (Jupyter Notebook)

Algorithm:

1. Store prices, quantities, discount rate & tax rate.
2. Subtotal = Sum of Price x quantity.
3. Discounted amount = Subtotal - (Subtotal x $\frac{\text{discount}}{100}$)
4. Total = discounted amount + (discounted amount x tax/100)
5. Display the total

CODE:

```
prices = [50,30,20,100]
quantities = [2,3,5,1]

discount_rate, tax_rate = 10,5

subtotal = sum(p*q for p,q in zip(prices, quantities))
discounted = subtotal - subtotal * discount_rate/100
total = discounted + discounted * tax_rate/100
print("Subtotal:", subtotal)
print("After discount:", discounted)
print("Total Cost:", total)
```

Input/Output :

Subtotal : 390
After discount : 351.0
Total Cost : 368.55

Result :

The total cost after discount and tax is Rs 368.55